

1. Swarm UAV Competition | TEKNOFEST

Competition Objective

- Develop **software and algorithms** for multiple UAVs operating as a coordinated **swarm**.
- Enable the UAV swarm to **complete designated tasks autonomously/cooperatively**.
- Demonstrate the developed algorithms using UAVs under **real-world conditions**.
- Encourage development and understanding of **Swarm UAV technology**.
- Explore potential **civilian and military applications**.

Who Can Participate?

- University **students and graduates** can participate.
- **Team size:** Minimum **3**, maximum **10** members.

Competition Evaluation Stages

The competition consists of **five main stages**:

1. **Preliminary Design Report**
2. **Preliminary Evaluation Presentation**
3. **Critical Design Report**
4. **Flight Evidence and Software Evaluation Video**
5. **Final Competition**

IMP DATES IN 2026 , 2027 NOT ANNOUNCED YET

Explanation	Date
Competition Application Deadline	15.03.2026
(PDR) Final Submission Date	27.03.2026 17:00 (GMT+3)
Announcement of PDR Results and Teams Advancing to the Next Round	01-04.04.2026

Question and Answer Meeting	10.04.2026
Preliminary Assessment Presentation (Simulation)	20-21.04.2026
Preliminary Assessment Presentation Announcement of Elimination Results	27-30.04.2026
Critical Design Report (CDR) Final Submission Date	19.06.2026 / 17.00 (GMT+3)
Announcement of CDR Results	06 - 10.07.2026
Flight Evidence and Software Architecture Video Final Submission Date	02.08.2026
Announcement Date for Video Results—Finalists	07.08.2026
Competition Date - Location	07 - 11.09.2026 - GAZIANTEP
TEKNOFEST Şanlıurfa	September 30 - October 4, 2026

Mission Flow

1. Autonomous Swarm Takeoff

- Minimum **3 UAVs**.
- All UAVs take off simultaneously.
- Only **one start command** is allowed.
- UAVs maintain the initial formation while climbing.

2. Navigate to QR Points

- Swarm autonomously travels between QR locations.
- Formation must be maintained during navigation.
- The formation rotates toward the direction of the next QR.
- UAV headings must adjust toward the travel direction.

3. QR Detection & Decoding

- At least **one UAV must have a camera**.

- At least one UAV detects and decodes the QR.
- The swarm selects the instructions corresponding to its **Team ID**.
- QR information determines the tasks and **next QR point**.

4. **Formation Change**

- Swarm must dynamically switch between requested formations.
- Required spacing between UAVs must be maintained.

5. **Pitch / Roll Maneuvers**

- Entire formation may be commanded to tilt around pitch or roll axes.
- Formation center remains fixed.
- After the maneuver, the swarm maintains the new angle until another formation/maneuver command changes it.

6. **Altitude Change**

- Entire swarm moves to the requested altitude.
- Formation must remain intact.

7. **Remove / Rejoin a UAV**

- QR specifies:
 - UAV ID to leave the swarm.
 - Red or blue landing zone.
- Selected UAV separates and lands in the specified zone.
- It disarms and waits.
- It must then re-arm and rejoin the swarm by the next QR task at the latest.
- Remaining UAVs must preserve their current formation.

8. **Colored Zone Detection**

- Swarm must detect **red and blue landing zones while flying the route**.
- Their positions must be recorded for later use.
- No additional scanning is allowed specifically to find them.

9. **Return Home**

- **Next QR = 0** means the final QR.
- Complete the QR task.
- Return autonomously to home.
- Land safely while maintaining formation.
- Disarm all UAVs.



2. TEKNOFEST Drone Championship | TEKNOFEST

No cz individual participation only no teams

World Drone Cup | TEKNOFEST

this one is also individual

international Unmanned Aerial Vehicle Competition | TEKNOFEST

Objective:

Develop students' UAV knowledge and technical skills, encourage innovative UAV projects, and bring together national and international students.

Participation:

Team-based competition for national and international associate, undergraduate, graduate, and eligible high-school students.

Categories:

- Fixed Wing
- Rotary Wing
- Free Mission

Evaluation:

- Project Presentation Report
- Preparation and Mission Video

Explanation	Date
Competition Deadline	March 15, 2026
Project Presentation Report Submission	March 9–16, 2026 - 17.00 (GMT+3)
Announcement of Project Presentation Report Results	April 13–17, 2026

Submission of Commitment and Consent Forms by Teams Successful in the Project Presentation Report April 21 – May 8, 2026

Deadline for Withdrawing from Competitions and Member Changes June 5, 2026

Submission of Preparation and Mission Videos July 13–17, 2026 - 17.00 (GMT+3)

Announcement of Preparation and Task Video Results August 10–14, 2026

Competition Week and Awards Ceremony September 8–13, 2026 / Malatya Tulga Havaalanı

TEKNOFEST TEKNOFEST Şanlıurfa September 30 – October 4, 2026

Fixed Wing	Rotary Wing	Free Mission	
Aircraft type	Fixed-wing only	Multirotor/helicopter (quad, hex, etc.)	Any type — fixed wing, VTOL, flapping wing, hybrid, rotary
Max weight	4 kg	4 kg	25 kg
Storage box shape	Cylindrical	Rectangular/square prism	None specified
Mission defined by	TÜBİTAK (fixed)	TÜBİTAK (fixed)	The team itself
# of missions	2 (Mission 1 + Mission 2)	2 mandatory + 1 optional bonus mission	1 self-designed mission, up to 3 flight attempts
VTOL allowed?	✗ No	✗ No	✓ Only here
Target detection (Int'l)	Image processing required	Image processing required	Team's choice

Target detection (High School)	Target location given (no image processing)	Image processing required (unlike Fixed Wing HS version)	Team's choice
Max flight time (M2)	10 min	10 min	15 min total

Competition Categories

1. **Fixed Wing** – autonomous fixed-wing aircraft
 2. **Rotary Wing** – multirotor/helicopter-type aircraft
 3. **Free Mission** – any UAV type (fixed wing, VTOL, flapping wing, hybrid, rotary), team defines its own mission
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MISSIONS

Fixed Wing Category (Int'l & Inter-High-School)

Mission 1 – unloaded flight test:

- Take off, fly a figure-eight (∞) maneuver around two poles (2 laps) staying outside the poles
- Scored on: storage-box volume, install/prep time, unloaded weight, lap time, autonomy level
- Fail this → can't attempt Mission 2

Mission 2 – payload drop:

- Same course, but now carrying 2 payloads (different weights, declared in advance)
- **International**: must *detect target locations via image processing* (targets' positions unknown) — red payload → 4×4 m blue target, blue payload → 2×2 m red target
- **Inter-High-School**: target locations are *given in advance* (no image processing needed) — blue payload → 10 m-diameter target, red payload → 2 m-diameter target
- Target-distance measured from where payload finally rests, up to 20 m from target center

Rotary Wing Category (Int'l & Inter-High-School)

Mission 1 – same figure-eight unloaded flight test as Fixed Wing, but with a rectangular/square storage box.

Mission 2 – payload drop, both competitions use image processing to find target locations:

- **International**: red payload → blue hexagon (2 m sides), blue payload → red triangle (1 m sides); measured up to 10 m from center

- **Inter-High-School:** red payload → 2×2 m blue target, blue payload → 1×1 m red target

Mission 3 (International only, optional) – after dropping both payloads, return to the first target, *pick up* that payload, and drop it on the *other* target — earns bonus points, not mandatory.

Free Mission Category

- No fixed mission — teams design their own scenario, described in the Project Presentation Report and Preparation/Mission Video
 - Same scenario must be reproduced exactly in the competition area
 - Max flight time: 15 minutes; 5-minute briefing to judges beforehand
 - Autonomous execution scores higher than manual
 - AAC can reject or require modification of a proposed mission on safety/area/rule grounds (e.g., paint gun instead of a real weapon mechanism, limits on high-speed aerobatics)
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KEY RESTRICTIONS

Team/eligibility

- Teams of 5–10 members; one Team Responsible required
- High-school students (Inter-High-School comp) or university/grad students (International); "mixed teams" allowed with <50% high schoolers
- One person can only be on one team, in one of the two competitions
- No first-degree kinship/conflict of interest between members and advisor

Aircraft build & weight

- Fixed Wing / Rotary Wing: max take-off weight **4 kg**; design/build must be done by the team (no ready-made or replica UAVs)
- Free Mission: max take-off weight **25 kg**; doesn't need to be original design, but parts must be declared
- VTOL UAVs are **barred** from Fixed Wing/Rotary Wing — Free Mission only

Safety-critical hardware

- Mandatory radio fail-safe (auto-triggers within 5 sec of signal loss)
- Mandatory emergency power cut-off switch (cuts power within 2 sec)
- End-to-end encrypted radio control and telemetry required
- Arm/disarm only via transmitter stick movements

Flight rules

- Max altitude: 120 m AGL
- Wind threshold for suspension: 25 km/h

- Only 1 camera allowed for image processing — a second observation camera is **strictly prohibited**
- No jamming, spoofing, or interference with other teams' signals (instant disqualification)
- Must use same motor/propeller across Mission 1 and Mission 2

Payload rules

- Payloads must be identical except color; only the specified paint color allowed (no QR codes, labels, stripes)
- No electronics inside payloads
- No parachute-type systems unless same color as payload
- Payloads must not break apart on impact (or scoring is voided)

Prohibited materials/content

- **No biological, chemical, or hazardous substances** on any flight
- **No explosions or anything risking human/environmental safety** — paint or bead-gun systems are suggested substitutes for "weapon-style" Free Mission concepts



3. FPV Drone Tracking Competition | TEKNOFEST

Competition Objective:

The primary aim is to evaluate the image-based perception, decision-making, and dynamic target tracking capabilities of developed autonomous aerial vehicles within a realistic mission scenario. The competition focuses on successfully tracking a moving target using image processing and autonomous control algorithms, particularly **without the use of GNSS (Global Navigation Satellite System)**. Teams are expected to develop systems that can autonomously track a moving target drone using vision alone, building capabilities in advanced target tracking, frame centering, lost-target reacquisition, and dynamic flight performance.

Who Can Participate:

- University students (associate degree, bachelor's, master's, doctoral) studying in Türkiye or abroad
- Graduates
- Team size: **4–10 members**, including the advisor

MISSION

Core task: Autonomously lock onto and track a moving target drone through a set of **11 predetermined maneuvers**, without using GNSS.

How it runs:

1. Team **manually** takes off and flies into the designated mission area (target drone starts stationary)
2. Once locked on, pilot switches **their own drone to autonomous mode**
3. Target drone then begins moving through a common, pre-set maneuver sequence
4. Competing drone must **autonomously track it** through all 11 maneuvers
5. Live camera feed from the tracking drone is shown to the referee throughout
6. At the end, pilot exits autonomous mode and returns the drone to the starting point

Timing:

- Team must reach competition area within **5 minutes** of being called
- **15 minutes** prep time
- **10 minutes max** task duration (timer starts when target drone takes off)
- If time runs out mid-task, score = however many maneuver stations were completed

Attempts: One attempt per day, 4 competition days total = **4 attempts**. Requesting a restart **resets all points** earned that attempt.

Scoring (Task = 75% of total, Reports/Video = 25%)

Task Step	Points
Live image transmission	10
Autonomous flight capability	10
Maneuver Station 1	10
Maneuver Station 2	10
Maneuver Station 3	15
Maneuver Station 4	20

	(Fixed Wing)	(Rotary Wing)	(Free Mission)	FPV Drone Tracking	Swarm UAV	Drone Championship	World Drone Cup
Objective	Autonomous flight + accurate payload drop	Same + optional pickup/relocate	Team-designed mission demo	Vision-based autonomous target tracking, no GNSS	Multi-UAV coordinated swarm behavior	Not detailed (racing-style, assumed)	Not detailed (racing-style, assumed)
Participation	Team	Team	Team	Team	Team	Individual only	Individual only
Who can join	High school + university/grad (mixed teams allowed)	Same	Same	University students + graduates	University students + graduates	Not specified	Not specified
Team size	5–10	5–10	5–10	4–10	3–10	N/A	N/A
Max weight	4 kg	4 kg	25 kg	UAV-1 class (4–25 kg)	Not specified per UAV	Unknown	Unknown
VTOL allowed?	✗	✗	✓ only here	Not applicable (fixed drone type)	Not specified	Unknown	Unknown
GNSS use	Allowed/expected	Allowed/expected	Allowed	Explicitly banned	Required for swarm nav	Unknown	Unknown
# missions/stages	2 missions	2 + 1 bonus	1 self-designed	1 task (11 maneuvers)	8-step mission flow (takeoff → return home)	Unknown	Unknown
Core skill tested	Precision drop + autonomy	Precision drop + retrieval	Creativity + engineering breadth	Real-time vision tracking w/o GPS	Multi-agent coordination, formation control	Piloting speed/skill	Piloting speed/skill

Evaluation stages	Report → Video → Live flight	Same	Same	Pre-Design → Critical Design → Video → Live flight	PDR → Sim presentation → CDR → Video → Live flight	Unknown	Unknown
Competition venue/date	Malatya (Sept 8–13) + Şanlıurfa TEKNOFEST	Same	Same	Şanlıurfa, Aug–Sept	Gaziantep (Sept 7–11) + Şanlıurfa	Unknown	Unknown
1st place prize	250,000 TRY	250,000 TRY	250,000 TRY	300,000 TRY	Not stated yet	Unknown	Unknown
Technical difficulty	High (custom airframe, no off-the-shelf parts)	High	Medium–High (open design)	High (vision-only tracking is a hard CV/controls problem)	Very high (multi-agent systems + comms)	Medium (piloting skill)	Medium (piloting skill)